

PAIRWISE EDGE CORRELATIONS IN RANDOM MINIMUM SPANNING TREES: A UNIVERSAL BOUND AND COMPLETE-GRAPH NEGATIVE CORRELATION

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ABSTRACT. Let G be a finite connected multigraph whose edges receive independent weights from one atomless law, and let $\text{MST}(G)$ be the resulting random minimum spanning tree. Its law is not pairwise negatively correlated: Lyons, Peres and Schramm exhibited two positively correlated edges, and we give such an example on a simple graph. We prove that positive correlation is nevertheless uniformly controlled:

$$\mathbf{P}_{\text{MST}}(e, f \in T) \leq 8 \mathbf{P}_{\text{MST}}(e \in T) \mathbf{P}_{\text{MST}}(f \in T),$$

answering a question of R. Lyons recorded by Tang and Zhang. After conditioning on all other weights, Harris's inequality gives conditional negative correlation; two bottleneck distances and a sharp second-moment estimate control the remaining environmental covariance. For K_n we prove pairwise negative correlation for every $n \geq 3$. The key finite identity is $\mathbf{E}[\deg(x)^2] = 10(n-1)/n - 4\mathbf{E}[L_n]$, where L_n is the total weight of the minimum spanning tree under rate-one exponential weights. Known expansions for $\mathbf{E}[L_n]$ then give the rate of convergence to $10 - 4\zeta(3)$ and the limits of both pair-correlation ratios. Finally, an explicit K_4 family shows that no universal constant survives when the independent edge laws need not be identical.

1. INTRODUCTION

Throughout, $G = (V, E)$ is a finite connected multigraph: loops and parallel edges are permitted. Each edge $g \in E$ receives a weight $w(g)$, and the weights are independent with one common atomless law μ on $\mathbb{R}$. Almost surely all weights are distinct, so the minimum spanning tree $T = \text{MST}(G)$ is almost surely unique; we write $\mathbf{P}_{\text{MST}}$ for its law, a probability measure on the spanning trees of G , equivalently on $\{0, 1\}^E$. Since the minimum spanning tree depends on the weights only through their relative order, $\mathbf{P}_{\text{MST}}$ does not depend on μ : uniform, exponential, or any other atomless law produce the same measure. We write

$$p_e = \mathbf{P}_{\text{MST}}(e \in T), \quad p_{ef} = \mathbf{P}_{\text{MST}}(e \in T, f \in T).$$

Uniform spanning trees have negatively correlated edge indicators, so it is natural to ask whether the minimum spanning tree law behaves the same way; for the uniform spanning tree measure the pairwise inequality $p_{ef} \leq p_e p_f$ is classical [14, Chapter 4]. Following [19], say that $\mathbf{P}_{\text{MST}}$ has the *pairwise negative correlation* (p-NC) property if $p_{ef} \leq p_e p_f$ for all $e \neq f$. The expectation is wrong. Lyons, Peres and Schramm [15, §5] take K_4 , replace each of two disjoint edges by three parallel copies, and mark one copy in each bundle; then

$$(1) \quad p_e = p_f = \frac{331}{1260}, \quad p_{ef} = \frac{109}{1575}, \quad \frac{p_{ef}}{p_e p_f} = \frac{109872}{109561} = 1.00283\dots > 1.$$

Thus the relevant quantity is the ratio $p_{ef}/(p_e p_f)$. We bound it uniformly over all graphs and determine its sign on the complete graph, where the naive inequality turns out to be true at every size.

Date: 7 August 2026.

2020 Mathematics Subject Classification. Primary 60C05; Secondary 05C80, 05C85.

Key words and phrases. minimum spanning tree, negative correlation, pair correlation, multiplicative coalescent, random graphs.

1.1. A universal factor. Tang and Zhang [19, Question 3.13] record the following question, which they attribute to R. Lyons: is there a constant $C > 0$ such that $p_{ef} \leq C p_e p_f$ for every finite connected graph G and every pair of distinct edges e, f of G ? Their motivation is the analogous inequality with $C = 2$ for the *uniform forest* measure, which follows from the work of Brändén and Huh on Lorentzian polynomials [5]; for further uniform spanning subgraph measures see [20].

Once the factor-1 inequality fails, a graph-independent multiplicative bound is a natural quantitative surrogate for negative dependence. The factor 2 for uniform forests shows that this surrogate remains meaningful for a neighbouring random spanning-subgraph measure; the question is whether any such control survives the change from a uniform measure to Kruskal’s order-driven law.

Theorem A. Let G be a finite connected multigraph whose edge weights are independent and identically distributed with an atomless law, and let $e \neq f$ be edges of G . Then

$$\mathbf{P}_{\text{MST}}(e, f \in T) \leq 8 \mathbf{P}_{\text{MST}}(e \in T) \mathbf{P}_{\text{MST}}(f \in T).$$

Theorem A is proved in §3. The multigraph setting includes the original obstruction (1); loops, bridges, parallel marked edges, and disconnected deleted environments are covered in §3.5.

The constant 8 is not optimal. Proposition 4.2 finds the exact maximum $78100/77841 = 1.00332\dots$ in the two-parameter parallel-bundle family containing (1), but simple graphs already do better: Proposition 4.4 gives the ratio $13938405/13872419 = 1.00475\dots$. Thus the optimal constant lies in $[13938405/13872419, 8]$. Corollary 3.5 improves the upper factor to $2 + o(1)$ when both marginals vanish.

Positive correlation is not an artefact of parallel edges. Proposition 4.3 gives a simple graph on five vertices with $p_{ef}/(p_e p_f) = 1450/1449 > 1$.

1.2. The complete graph, at every size. For K_n with $n \geq 3$ every edge has the same inclusion probability

$$(2) \quad p_0 = \mathbf{P}(e \in \text{MST}(K_n)) = \frac{n-1}{\binom{n}{2}} = \frac{2}{n},$$

because every spanning tree has $n-1$ edges and the edges of K_n are equivalent under automorphisms. Write p_1 for the probability that two fixed edges sharing a vertex both lie in the tree, and p_2 for the same probability for two fixed disjoint edges (defined for $n \geq 4$).

Tang and Zhang [19, Theorem 1.1] prove that $p_2 \leq p_0^2$ for all n beyond an unspecified threshold, by running Fatou’s lemma against the local weak limit of $\text{MST}(K_n)$ [1]; their argument yields no value for the threshold. For adjacent pairs they reduce the problem to an upper bound on $\mathbf{E}[\deg(x)^2]$ and leave it open, writing that while the matching lower bound “via Fatou’s lemma and the local limit . . . is straightforward, proving a corresponding upper bound seems more challenging” [19, §3.2.2]. Both cases in fact hold at every size, strictly.

Theorem B1. For every $n \geq 3$ and every two distinct edges of K_n sharing a vertex,

$$p_1 < p_0^2 = \frac{4}{n^2}.$$

Theorem B2. For every $n \geq 4$ and every two disjoint edges of K_n ,

$$p_2 < p_0^2 = \frac{4}{n^2}.$$

Corollary B3. For every $n \geq 3$ the measure $\mathbf{P}_{\text{MST}}$ on K_n has the full pairwise negative correlation property. (For $n = 3$ there is no disjoint pair, so the statement there is exactly Theorem B1.)

These are proved in §5. Theorem B1 settles the adjacent case left open in [19, §3.2.2]. Theorem B2 gives an effective complete-range proof of the disjoint conclusion of [19, Theorem 1.1], by a different method and without local weak convergence or Fatou's lemma. The two inequalities are strict. In the computed range, p_1/p_0^2 increases from 0.7500 at $n = 3$ to 0.7851 at $n = 30$, and p_2/p_0^2 from 0.9778 at $n = 4$ to 0.9948 at $n = 30$ (Table 1). Strictness is a feature of K_n and not of the measure in general: if e and f are two bridges of a connected graph then $p_e = p_f = p_{ef} = 1$, and $p_{ef} = p_e p_f$ exactly.

1.3. An exact identity for the complete graph. Theorems B1 and B2 are two sides of one estimate, and the object they estimate is classical. Let L_n denote the total weight of $\text{MST}(K_n)$ when the edge weights are i.i.d. $\text{Exp}(1)$.

Proposition D1. For every $n \geq 2$ and every vertex x of K_n ,

$$\mathbf{E}_{\text{MST}(K_n)}[\deg(x)^2] = \frac{10(n-1)}{n} - 4\mathbf{E}[L_n].$$

Corollary D2. For $n \geq 3$, adjacent pairwise negative correlation on K_n is equivalent to the first estimate below; for $n \geq 4$, disjoint pairwise negative correlation is equivalent to the second:

$$\begin{aligned} p_1 \leq p_0^2 &\iff \mathbf{E}[L_n] \geq \frac{(n-1)(n+2)}{n^2} \quad (\rightarrow 1), \\ p_2 \leq p_0^2 &\iff \mathbf{E}[L_n] \leq \frac{(n-1)(5n+6)}{4n^2} \quad \left(\rightarrow \frac{5}{4}\right). \end{aligned}$$

Both estimates themselves hold for every $n \geq 2$. The first is strict for $n \geq 3$, while the second is strict for $n \geq 4$. Moreover, with $H_n = \sum_{k=1}^n 1/k$, the lower estimate strengthens to

$$\mathbf{E}[L_n] \geq 1 + \frac{H_n - 1}{n} - \frac{1}{n^2}.$$

This strengthened bound is exact for $n = 2, 3$ and strict for $n \geq 4$.

Corollary D3. $\lim_{n \rightarrow \infty} \mathbf{E}_{\text{MST}(K_n)}[\deg(x)^2] = 10 - 4\zeta(3) = 5.1917723873616\dots$, which answers [19, Question 3.8] affirmatively and identifies the limit with the second moment $\mathbf{E}[N^2]$ of the root degree of the wired minimal spanning forest on the Poisson-weighted infinite tree computed in [19, Proposition 3.11].

Thus one merger functional controls both pair correlations and the classical MST weight. Proposition D1 also explains the constant $10 - 4\zeta(3)$ in the PWIT calculation of [19]. Only Corollary D3 and its rate use the asymptotic results of [6, 13]; Theorems B1 and B2 remain finite and effective at every n .

1.4. Identically distributed weights. Question 3.13 is posed for a single common weight law. Since $\mathbf{P}_{\text{MST}}$ itself does not depend on which atomless law is used, one might expect that hypothesis to be cosmetic. It is not: as soon as different edges may carry different laws, no universal constant exists, already on K_4 .

Theorem C. For each integer $t \geq 1$ let G_t be the complete graph on $\{a, b, c, d\}$ in which the two disjoint edges ab and cd carry the law with distribution function x^t on $[0, 1]$ (that is, the law of $\max(U_1, \dots, U_t)$) and the four remaining edges are uniform on $[0, 1]$, all six weights independent. Write $R(t)$ for the ratio $p_{ab,cd}/(p_{ab}p_{cd})$ of the resulting minimum spanning tree measure. Then

$$R(t) = \frac{2(5t+6)(t+1)(t+4)^2(2t+3)}{9(5t^2+12t+8)^2}, \quad R(t) = \frac{4t}{45} + \frac{46}{75} + O(t^{-1}),$$

and $R(t)$ is strictly increasing on the positive integers, first exceeding 8 at $t = 84$. In particular $R(t) \rightarrow \infty$, so no inequality of the form of Question 3.13 holds, with any constant, for independent non-identically distributed atomless edge weights.

Subdividing an edge into a path of t i.i.d. uniform edges realizes the same maximum-weight law but not the same marked-edge event: requiring one fixed subdivided edge to lie in the tree introduces an attenuation that destroys the amplification above (Remark 4.5). Thus Theorem C does not contradict the i.i.d. theorem.

1.5. The idea of the proofs.

Theorem A. The obstruction (1) rules out a universal factor 1. Freeze the weights of all edges other than e and f — call this the *environment* — and let a, b, c be the conditional probabilities of $\{e \in T\}$, $\{f \in T\}$ and $\{e, f \in T\}$ given it, so that these are functions of the environment alone. By Kruskal’s rule the edge e is accepted exactly when its two endpoints are still separated by the strictly lighter edges. At frozen environment this makes the indicator of $\{e \in T\}$ nonincreasing in $w(e)$ and nondecreasing in $w(f)$, and symmetrically for f : raising the weight of one marked edge helps the other. Two functions of two independent coordinates that are monotone in opposite directions have nonpositive covariance, so Harris’s inequality gives $c \leq ab$ pointwise, with no further hypothesis on the tie-free environment. Conditional negative correlation therefore holds almost surely, and the whole failure exhibited by (1) lives in the covariance of a and b under the environment measure.

The remaining term is the environmental covariance. Its natural proxy is the bottleneck distance A between the endpoints of e in the environment graph $H = G - \{e, f\}$: were f absent, e would be accepted precisely when $w(e) < A$, so a would equal A exactly. The presence of f perturbs this by at most a factor 2 in each direction (Lemma 3.4), so with B the bottleneck distance between the endpoints of f it suffices to bound $\mathbf{E}[AB] \leq \sqrt{\mathbf{E}[A^2]\mathbf{E}[B^2]}$, Cauchy–Schwarz absorbing whatever dependence A and B have. What remains is to compare $\mathbf{E}[A^2]$ with $(\mathbf{E}A)^2$, and here the probabilistic content is that A is a percolation first-passage time: $\mathbf{P}(A > t)$ is the probability that two vertices are not connected in Bernoulli(t) percolation on H , and realising density $s + t$ as two independent sprinkled layers shows that this tail is submultiplicative. A nonnegative random variable with submultiplicative tail — a new-better-than-used random variable — satisfies $\mathbf{E}[A^2] \leq 2(\mathbf{E}A)^2$. Three factors of 2, one from that moment bound and one from each of the two perturbation estimates, produce the constant 8.

The complete graph. Here the mechanism is different and the answer is exact. Give every edge of K_n an independent rate-one exponential clock, run Kruskal in the order in which the clocks ring, and record only the accepted mergers. The resulting process on partitions of the vertex set is the discrete skeleton of the multiplicative coalescent: from components of sizes $a_1, \dots, a_k$ the next accepted edge joins two of them with probability proportional to $a_i a_j$, and its two endpoints are uniform in their components, independently of the past. Uniformity of the endpoints is what makes the degree functional computable: if $\Phi = \sum_x \deg(x)(\deg(x) - 1)$ counts ordered pairs of adjacent tree edges, then a merger of blocks of sizes a, b raises $\mathbf{E}\Phi$ by $8 - 4/a - 4/b$, so that $\mathbf{E}[\Phi \mid \text{merger history}] = 8(n - 1) - 4H$ with H the sum of $1/a + 1/b$ over the $n - 1$ mergers. Both pair probabilities on K_n are therefore governed by the single functional H , adjacent negative correlation asking for it to be large and disjoint negative correlation for it to be small. A pathwise identity — every component other than the last one is the child of exactly one merger — rewrites H as $n - \frac{1}{n} + J$, where J sums $1/|C|$ over the components C created by mergers. Chronology gives the sharper pathwise estimate $J \geq H_n - 1$: the component created by the j th merger has at most $j + 1$ vertices. This is Theorem B1. The other direction needs J bounded above, no pathwise bound is available (Remark 5.8), and this is where the multiplicative merger law enters: a one-step convexity inequality bounds the multiplicatively weighted average of $1/(a_i + a_j)$ over pairs of blocks by $k/(2n)$, and summing over the $n - 1$ states visited gives $\mathbf{E}[J] \leq (n - 1)(n + 2)/(4n)$, which is Theorem B2. Finally, in continuous time H accumulates

at rate $n(\kappa - 1)$, where κ is the number of components, while $L_n = \int_0^\infty (\kappa(t) - 1) dt$; so $\mathbf{E}[H] = n \mathbf{E}[L_n]$, which is Proposition D1.

2. PRELIMINARIES

2.1. The cycle characterisation. We use one description of the minimum spanning tree throughout. For a real u let $G_{<u}$ denote the subgraph of G consisting of the edges of weight strictly less than u .

Lemma 2.1 (Kruskal / cycle property). *Suppose all edge weights of G are distinct. Then for every edge g with endpoints x, y ,*

$$g \in \text{MST}(G) \iff x \text{ and } y \text{ lie in different components of } G_{<w(g)}.$$

In particular a loop is never in $\text{MST}(G)$, and a bridge always is.

Proof. This is the standard correctness statement for Kruskal's algorithm, which processes edges in increasing weight and accepts exactly those joining two distinct current components; with distinct weights the algorithm's output is the unique minimum spanning tree, and the set of edges present when g is processed is exactly $E(G_{<w(g)})$. For a loop, $x = y$ always lie in the same component. For a bridge g , $G_{<w(g)} \subseteq G - g$ never connects x to y . $\square$

Since $\text{MST}(G)$ depends on the weight vector only through its order, and since applying the common distribution function F_μ to every weight is an almost surely strictly order-preserving map onto independent $\text{Uniform}[0, 1]$ weights, we may and do assume in §3 that all weights are uniform on $[0, 1]$. In §5 we instead use $\text{Exp}(1)$ weights, again without changing $\mathbf{P}_{\text{MST}}$.

2.2. Connection times.

Definition 2.2. Let K be a finite multigraph with weights in $[0, 1]$ and let x, y be vertices. Set

$$T_K(x, y) = \inf\{t \in [0, 1] : x \leftrightarrow y \text{ in } K_{<t}\},$$

with the convention $T_K(x, y) = 1$ if x and y lie in different components of K , and $T_K(x, x) = 0$.

Equivalently $T_K(x, y)$ is the bottleneck distance from x to y , the minimum over paths of the largest weight on the path (and 1 if there is no path). Definition 2.2 is the threshold relevant to adding a fresh edge: by Lemma 2.1, if a new edge xy of weight u is added to K then, outside the null set of ties, it is accepted precisely when $u < T_K(x, y)$.

3. A UNIVERSAL FACTOR FOR ALL FINITE MULTIGRAPHS

Throughout this section the weights are i.i.d. $\text{Uniform}[0, 1]$. We begin with the estimate that ultimately pays for the correlation, and which is the only place where a percolation argument is used.

3.1. A second-moment bound for connection times.

Lemma 3.1. *Let K be a finite multigraph with i.i.d. uniform weights, let x, y be vertices, put $T = T_K(x, y)$ and $q(t) = \mathbf{P}(T > t)$, extended by $q(t) = 0$ for $t \geq 1$. Then*

$$(3) \quad q(s+t) \leq q(s)q(t) \quad (s, t \geq 0), \quad \text{and consequently} \quad \mathbf{E}[T^2] \leq 2(\mathbf{E}T)^2.$$

Proof. For $s+t \geq 1$ the left side vanishes, so assume $s+t < 1$. For $u \in [0, 1]$ the subgraph $K_{<u}$ is $\text{Bernoulli}(u)$ percolation on $E(K)$, and $q(u)$ is the probability that x and y are not connected in it. Realise $\text{Bernoulli}(s+t)$ percolation as the union of two independent layers of densities s and $r = t/(1-s)$: an edge is absent from the union with probability $(1-s)(1-r) = 1-s-t$, as required. If the union fails to connect x to y , then each layer

separately fails to connect them. The two layers are independent, q is nonincreasing, and $r \geq t$; hence

$$q(s+t) \leq q(s)q(r) \leq q(s)q(t).$$

This also covers the case in which x, y lie in different components of K , where $q \equiv 1$ on $[0, 1)$. For the second assertion, all integrals below are over $[0, \infty)$ and q vanishes on $[1, \infty)$, so by Tonelli

$$(\mathbf{E}T)^2 = \iint q(s)q(t) ds dt \geq \iint q(s+t) ds dt = \int u q(u) du = \frac{1}{2} \mathbf{E}[T^2]. \quad \square$$

The submultiplicativity in (3) is the two-layer sprinkling decomposition behind the classical square-root trick, and its consequence is the standard second-moment inequality for a new-better-than-used random variable; see e.g. [3].

Remark 3.2. The constant 2 in (3) cannot be improved. If K consists of k parallel xy -edges then T is the minimum of k uniforms, so $\mathbf{E}T = 1/(k+1)$, $\mathbf{E}[T^2] = 2/((k+1)(k+2))$ and

$$\frac{\mathbf{E}[T^2]}{(\mathbf{E}T)^2} = \frac{2(k+1)}{k+2} \nearrow 2.$$

3.2. Freezing the environment. Fix distinct edges $e \neq f$ of G , write $H = G - \{e, f\}$ for the multigraph on the same vertex set with e and f deleted, and put $U = w(e)$, $V = w(f)$. Let $\mathcal{F} = \sigma(w(g) : g \in E(H))$ be the σ -field generated by the *environment*, i.e. by all weights other than those of the two marked edges. For a tie-free realisation W of the environment, which occurs almost surely, set

$$(4) \quad a(W) = \mathbf{P}(e \in T \mid \mathcal{F} = W), \quad b(W) = \mathbf{P}(f \in T \mid \mathcal{F} = W), \quad c(W) = \mathbf{P}(e, f \in T \mid \mathcal{F} = W),$$

where the conditional probabilities average only over the independent uniform pair (U, V) .

Lemma 3.3 (conditional negative correlation). *For every tie-free environment W ,*

$$c(W) \leq a(W)b(W).$$

Proof. Fix W and regard the acceptance indicators $\mathbf{1}_{\{e \in T\}}$ and $\mathbf{1}_{\{f \in T\}}$ as functions of $(U, V) \in [0, 1]^2$. By Lemma 2.1, e is accepted precisely when its endpoints are not connected by the edges of weight $< U$, that is by

$$\{g \in E(H) : w(g) < U\} \cup \{f \text{ if } V < U\}.$$

Increasing U only enlarges this edge set, so $\mathbf{1}_{\{e \in T\}}$ is nonincreasing in U ; increasing V can only remove f from it, so $\mathbf{1}_{\{e \in T\}}$ is nondecreasing in V . By symmetry $\mathbf{1}_{\{f \in T\}}$ is nondecreasing in U and nonincreasing in V .

Now change coordinates to $(U, 1 - V)$, which are again independent and uniform. In these coordinates $\mathbf{1}_{\{e \in T\}}$ is coordinatewise nonincreasing and $\mathbf{1}_{\{f \in T\}}$ is coordinatewise nondecreasing. Harris's inequality [11] on a product of two totally ordered probability spaces gives nonpositive covariance for a decreasing and an increasing function, which is exactly $c(W) \leq a(W)b(W)$. $\square$

The conditioning is essential: by (1), the unconditional failure lives in the covariance of the two functions a and b . We compare them with connection times in the graph with both marked edges deleted.

3.3. A deletion sandwich.

Lemma 3.4. *Let e have endpoints x_1, x_2 and f have endpoints y_1, y_2 , and set*

$$A = A(W) = T_H(x_1, x_2), \quad B = B(W) = T_H(y_1, y_2)$$

for the connection times in the deleted environment H (Definition 2.2). Then, for every tie-free W ,

$$\frac{A}{2} \leq A - \frac{A^2}{2} \leq a(W) \leq A, \quad \frac{B}{2} \leq B - \frac{B^2}{2} \leq b(W) \leq B.$$

Proof. Upper bound. Adding f to H can only connect x_1 to x_2 at a smaller or equal threshold, so if e is accepted then, outside the null set $\{U = A\}$, we have $U < A$; the set $\{U < A\} \subseteq [0, 1]^2$ has area A .

Lower bound. If $U < \min(A, V)$ then f is not among the edges of weight $< U$, and H alone does not connect x_1 to x_2 below U because $U < A$; so e is accepted by Lemma 2.1. The area of $\{U < \min(A, V)\}$ is

$$\int_0^1 \min(A, v) dv = \frac{A^2}{2} + A(1 - A) = A - \frac{A^2}{2} \geq \frac{A}{2},$$

the last step because $A \leq 1$. The statements for f follow by exchanging the roles of e and f . $\square$

3.4. Proof of Theorem A. We combine conditional negative correlation, the deletion sandwich, and the connection-time moment bound.

Proof of Theorem A. By §2 we may take the weights uniform on $[0, 1]$. With the notation of (4) and Lemma 3.4, and taking all expectations with respect to the environment,

$$\mathbf{P}_{\text{MST}}(e, f \in T) = \mathbf{E}[c] \stackrel{\text{L.3.3}}{\leq} \mathbf{E}[ab] \stackrel{\text{L.3.4}}{\leq} \mathbf{E}[AB] \leq \sqrt{\mathbf{E}[A^2] \mathbf{E}[B^2]} \stackrel{\text{L.3.1}}{\leq} 2\mathbf{E}[A] \mathbf{E}[B],$$

where the third step is Cauchy–Schwarz and the fourth applies Lemma 3.1 separately to the two terminal pairs $\{x_1, x_2\}$ and $\{y_1, y_2\}$ in the multigraph H with its i.i.d. uniform weights. Finally Lemma 3.4 gives $\mathbf{E}[A] \leq 2\mathbf{E}[a] = 2\mathbf{P}_{\text{MST}}(e \in T)$ and likewise $\mathbf{E}[B] \leq 2\mathbf{P}_{\text{MST}}(f \in T)$, so

$$\mathbf{P}_{\text{MST}}(e, f \in T) \leq 2 \cdot 2\mathbf{P}_{\text{MST}}(e \in T) \cdot 2\mathbf{P}_{\text{MST}}(f \in T) = 8\mathbf{P}_{\text{MST}}(e \in T) \mathbf{P}_{\text{MST}}(f \in T). \quad \square$$

The same estimates improve the constant when both marked edges are rare.

Corollary 3.5 (small marginals). *Put $g(p) = (1 - \sqrt{1 - 4p})/2$. If $p_e, p_f \leq 1/4$, then*

$$p_{ef} \leq 2g(p_e)g(p_f).$$

If in addition $p_e p_f > 0$, then

$$\frac{p_{ef}}{p_e p_f} \leq 2 \frac{g(p_e)}{p_e} \frac{g(p_f)}{p_f}.$$

Consequently the upper bound on the ratio is $2 + o(1)$ as $\max(p_e, p_f) \rightarrow 0$.

Proof. Write $m_A = \mathbf{E}[A]$ and $m_B = \mathbf{E}[B]$. Lemmas 3.4 and 3.1 give

$$p_e = \mathbf{E}[a] \geq m_A - \frac{1}{2}\mathbf{E}[A^2] \geq m_A - m_A^2, \quad p_e \geq \frac{1}{2}m_A,$$

and likewise for f . If $p_e \leq 1/4$, the second inequality gives $m_A \leq 1/2$, and inversion of the increasing function $m \mapsto m - m^2$ on $[0, 1/2]$ gives $m_A \leq g(p_e)$. The proof of Theorem A already gives $p_{ef} \leq 2m_A m_B$, which proves the assertions; finally $g(p)/p \rightarrow 1$ as $p \downarrow 0$. $\square$

3.5. Degenerate configurations. No simplicity or 2-connectivity hypothesis has entered, and the degenerate cases are covered rather than excluded. If e is a loop then $x_1 = x_2$, so $A = 0$, $a \equiv 0$ and $p_e = p_{ef} = 0$, and the inequality reads $0 \leq 0$. If e is a bridge of G then H does not connect x_1 to x_2 , so $A = 1$ and $p_e = 1$; the chain above is valid, and one should expect no strictness, since two bridges give $p_{ef} = p_e p_f = 1$. If e and f have the same endpoints then $p_{ef} = 0$, because the heavier of the two closes a cycle with the lighter, and Lemma 3.3 remains valid. Finally $H = G - \{e, f\}$ need not be connected; Definition 2.2 and Lemma 3.1 were set up to cover this ($T = 1$), and Lemma 3.4 is pointwise, so nothing changes.

4. EXPLICIT CORRELATION EXAMPLES

We first quantify the parallel-bundle construction of [15], then give a simple-graph counterexample to pairwise negative correlation, and finally show why the common-law hypothesis of Theorem A is necessary. The two K_4 families use the following four-terminal calculation.

4.1. A four-terminal formula. Label the vertices of K_4 by a, b, c, d and distinguish the two disjoint edges

$$A = ab, \quad B = cd.$$

The remaining four edges ac, ad, bc, bd form a complete bipartite graph between $\{a, b\}$ and $\{c, d\}$; give them a common continuous distribution function H , and give A, B the continuous distribution functions F, G , all six weights independent. Write $\bar{H} = 1 - H$.

Lemma 4.1. *With the above notation,*

$$(5) \quad p_A = \int \left[(1 - G(x))(1 - H(x)^2)^2 + G(x)\{2\bar{H}(x)^2 - \bar{H}(x)^4\} \right] dF(x),$$

$$(6) \quad p_B = \text{the same with } (F, A) \text{ and } (G, B) \text{ interchanged,}$$

$$(7) \quad p_{AB} = \iint_{x < y} \bar{H}(y)^2 [2(1 - H(x)^2) - \bar{H}(y)^2] dF(x) dG(y) \\ + \iint_{y < x} \bar{H}(x)^2 [2(1 - H(y)^2) - \bar{H}(x)^2] dF(x) dG(y).$$

Proof. For (5), condition on $w(A) = x$; by Lemma 2.1, A is accepted iff $a \not\leftrightarrow b$ below x .

If $w(B) > x$, which has probability $1 - G(x)$, only cross edges are present below x , and $a \leftrightarrow b$ below x iff c receives both of ac, bc below x , or d receives both of ad, bd below x . These two events are independent, each of probability $H(x)^2$, so the probability of non-connection is $(1 - H(x)^2)^2$.

If $w(B) < x$, of probability $G(x)$, then c and d are already merged, and $a \leftrightarrow b$ below x iff a has at least one cross edge below x and b has at least one; those events are independent with probability $1 - \bar{H}(x)^2$ each. The probability of non-connection is $1 - (1 - \bar{H}(x)^2)^2 = 2\bar{H}(x)^2 - \bar{H}(x)^4$.

For (7), take the ordering $x = w(A) < y = w(B)$ and write $q = 1 - H(x)^2$, $v^2 = \bar{H}(y)^2$. For each $u \in \{c, d\}$ let N_u be the event that u has no cross edge below y (probability v^2) and B_u the event that u does not receive both of its cross edges below x (probability q); $N_u \subseteq B_u$ since $x < y$, and the pairs of events at c and at d are independent. As above, A is accepted iff $B_c \cap B_d$ occurs. Given that A has been accepted and lies below y , the vertices a, b are merged, and B is accepted iff at most one of c, d is attached to that merged block below y , i.e. iff $N_c \cup N_d$ occurs. Hence

$$\mathbf{P}(A, B \text{ accepted} \mid x, y) = \mathbf{P}((B_c \cap B_d) \cap (N_c \cup N_d)) = v^4 + 2v^2(q - v^2) = 2qv^2 - v^4,$$

which is the first integrand of (7). The other order is symmetric. $\square$

4.2. Parallel bundles, and a lower bound for the optimal constant. Enlarging the two parallel bundles in the construction of [15] does not increase the ratio indefinitely; the exact maximum is finite. Let $R_{r,s}$ denote the ratio for one marked copy in each of two disjoint parallel bundles of sizes r, s on K_4 , the four cross edges uniform. Regard each bundle as a macro-edge whose weight is its minimum. If a macro-edge is selected, its unique lightest copy is selected, and symmetry therefore divides its marginal by r or s and the joint probability by rs ; these factors cancel in the ratio. Substituting the laws of $\min(U_1, \dots, U_r)$ and $\min(V_1, \dots, V_s)$ into Lemma 4.1, and then applying these symmetry factors, gives, writing $D = (r + s + 2)(r + s + 3)(r + s + 4)$ and $I(r, s) = \frac{4}{(s+2)(r+s+3)} - \frac{3s+10}{(s+2)(s+4)(r+s+4)}$,

$$p_e = \frac{r+6}{(r+2)(r+4)} + \frac{4}{D}, \quad p_f = \frac{s+6}{(s+2)(s+4)} + \frac{4}{D}, \quad p_{ef} = I(r, s) + I(s, r).$$

Proposition 4.2. *For all positive integers r, s ,*

$$R_{r,s} = \frac{p_{ef}}{p_e p_f} \leq \frac{78100}{77841} = 1.003327\dots,$$

with equality exactly at $r = s = 4$. In the symmetric subfamily,

$$R_{t,t} = \frac{(t+1)^2(t+4)(2t+3)(2t^2+19t+34)}{(2t^3+17t^2+34t+22)^2},$$

$$R_{t,t} - 1 = \frac{20t^3+9t^2-78t-76}{(2t^3+17t^2+34t+22)^2} = \frac{5}{t^3} + O(t^{-4}),$$

so the ratio peaks at $t = 4$ and returns to 1. The configuration of [15] is $(r, s) = (3, 3)$, with $R_{3,3} = 109872/109561$.

Proof. Put $m = r + s$ and $\rho = rs$, and define

$$\begin{aligned} D &= (m+2)(m+3)(m+4), & K &= (m+2)^2(m+3)(m+4), \\ L &= D + 24, & C &= 6D + 32, \\ N_0 &= K(6m^3 + 78m^2 + 356m + 544), & N_1 &= K(m^2 + 7m + 16), \\ D_0 &= 4Cm^2 + LCm + C^2, & D_1 &= L^2 + 4Lm - 8C. \end{aligned}$$

Simplifying the three displayed probabilities gives

$$(8) \quad R_{r,s} = \frac{N_0 + N_1\rho}{D_0 + D_1\rho + 16\rho^2}.$$

For fixed m , the derivative with respect to ρ has the sign of

$$Q_m(\rho) = N_1D_0 - N_0D_1 - 32N_0\rho - 16N_1\rho^2,$$

and $Q'_m(\rho) = -32N_0 - 32N_1\rho < 0$ for $\rho \geq 0$. The feasible range is $m-1 \leq \rho \leq \lfloor m^2/4 \rfloor$.

For $2 \leq m \leq 10$, direct substitution at the right endpoint gives

$$\begin{aligned} (Q_m(\lfloor m^2/4 \rfloor))_{m=2}^{10} &= (623232000, 3726475200, 15482880000, 50485284864, \\ &\quad 133754572800, 297101597760, 541023436800, \\ &\quad 751512453120, 464792380416), \end{aligned}$$

so R is increasing throughout the feasible interval. Its balanced endpoint values, in the same order, are

$$\frac{44}{45}, \frac{3745}{3774}, \frac{840}{841}, \frac{14385}{14362}, \frac{109872}{109561}, \frac{763587}{761105}, \frac{78100}{77841}, \frac{385385}{384154}, \frac{83772}{83521}.$$

The largest is the $m = 8$ value.

For $m \geq 11$, put $u = m - 11$. At the left endpoint,

$$\begin{aligned} -Q_m(m-1) &= 930527357760 + 3158322926304u + 1989328507888u^2 \\ &\quad + 609067397832u^3 + 112783678672u^4 + 13814498848u^5 \\ &\quad + 1164466368u^6 + 68281008u^7 + 2747600u^8 + 72576u^9 \\ &\quad + 1136u^{10} + 8u^{11} > 0. \end{aligned}$$

Thus R is decreasing on the feasible interval. If Num and Den denote the numerator and denominator of (8) at $\rho = m-1$, then

$$\begin{aligned} 78100 \text{ Den} - 77841 \text{ Num} &= 34642293840 + 28712778516u + 9667374224u^2 \\ &\quad + 1607247273u^3 + 144551244u^4 + 7147790u^5 \\ &\quad + 181300u^6 + 1813u^7 > 0. \end{aligned}$$

This proves the bound for $m \geq 11$ as well. Equality in the full argument requires $m = 8$ and $\rho = 16$, hence $r = s = 4$. Substitution of $r = s = t$ in (8) gives the two symmetric formulas, and $t = 3$ gives the values from [15]. $\square$

Thus the parallel-bundle route cannot push the ratio above 1.0034. The simple graphs below exceed this family maximum. Appendix A records the exact certificate checks.

4.3. Positive correlation in a simple graph. Parallel edges are not necessary for the failure of pairwise negative correlation.

Proposition 4.3. *Let G be the simple graph on $\{a, b, c, d, z\}$ with*

$$E(G) = \{ab, ac, ad, az, bc, bd, bz, cd\},$$

and mark $e = az$ and $f = cd$. Under i.i.d. atomless edge weights,

$$p_e = \frac{7}{12}, \quad p_f = \frac{69}{140}, \quad p_{ef} = \frac{145}{504}, \quad \frac{p_{ef}}{p_e p_f} = \frac{1450}{1449} > 1.$$

Proof. All $8!$ strict edge orders are equally likely. Running Kruskal's rule through those orders accepts e in 23520 orders, accepts f in 19872 orders, and accepts both in 11600 orders. Dividing by $8! = 40320$ gives the three stated probabilities and hence the ratio. Exact summation by relative-order patterns gives the same values. $\square$

This witness is minimal by edge count. Indeed, a smallest counterexample may be assumed bridgeless: a bridge belongs to every spanning tree, and deleting all bridges separates independent minimum-spanning-tree problems. A connected bridgeless simple graph with m edges has at most m vertices. An exhaustive exact census of all such graphs with $m \leq 7$ finds no violation.

The smallest witness is not the strongest one we know. Let S be a set of hub vertices with no internal edges, take four further vertices a, b, c, d , and join every vertex of S to all four of a, b, c, d . Add the two marked edges $A = ab$ and $B = cd$.

Proposition 4.4. *For the preceding simple graph, two hubs give*

$$p_A = p_B = \frac{1}{2}, \quad p_{AB} = \frac{1186}{4725}, \quad \frac{p_{AB}}{p_A p_B} = \frac{4744}{4725}.$$

Three hubs give

$$p_A = p_B = \frac{1123}{2730}, \quad p_{AB} = \frac{71479}{420420}, \quad \frac{p_{AB}}{p_A p_B} = \frac{13938405}{13872419} = 1.004756 \dots$$

In particular the second ratio is larger than the maximum $78100/77841$ in Proposition 4.2.

For the finite range $1 \leq |S| \leq 9$, exact evaluation shows that the ratio is maximized at $|S| = 3$ and is below 1 for $|S| = 7, 8, 9$. No assertion is made for all hub counts.

Proof. For two hubs, full enumeration of the $10!$ strict edge orders gives marginal counts 1814400 and joint count 910848. For three hubs, exact summation over the relative positions of the two marked edges and the other twelve edges gives marginal count 35861253120 and joint count 14821885440 out of $14!$ orders. Dividing gives the stated probabilities. The relative-position calculation is also applied to the two-hub graph and agrees with its full enumeration. For $1 \leq |S| \leq 9$, grouping the relative-order patterns by the two induced partitions of $\{a, b, c, d\}$ gives the finite-range comparison exactly. $\square$

4.4. Non-identical laws: proof of Theorem C. The second specialisation replaces the parallel bundles, which are a device for producing a min of uniforms, by their opposite.

Proof of Theorem C. Take $F(x) = G(x) = x^t$ and $H(x) = x$ on $[0, 1]$ in Lemma 4.1. All integrands are then polynomials and the integrals are elementary; carrying them out gives (9)

$$p_A = p_B = A_t := \frac{3(5t^2 + 12t + 8)}{(t+1)(t+2)(t+4)(2t+3)}, \quad p_{AB} = J_t := \frac{2(5t+6)}{(t+1)(t+2)^2(2t+3)}.$$

Therefore

$$R(t) = \frac{J_t}{A_t^2} = \frac{2(5t+6)(t+1)(t+4)^2(2t+3)}{9(5t^2 + 12t + 8)^2},$$

a rational function whose numerator has degree 5 and whose denominator has degree 4, with leading coefficients 20 and 225. Hence $R(t) \rightarrow \infty$ and $R(t)/t \rightarrow 20/225 = 4/45$; expanding one order further gives the constant $46/75$.

For monotonicity write $R(t) = 2N(t)/(9D(t))$, omitting the displayed constant factors from the numerator and denominator. The sign of $R(t+1) - R(t)$ is the sign of $Q(t) = N(t+1)D(t) - N(t)D(t+1)$. With $u = t - 1$, exact expansion gives

$$\begin{aligned} Q(u+1) = & 124000 + 669300u + 1289264u^2 + 1262269u^3 + 712831u^4 \\ & + 242311u^5 + 48935u^6 + 5400u^7 + 250u^8, \end{aligned}$$

which is positive for $u \geq 0$. Direct substitution gives $R(83) < 8 < R(84)$, so 84 is the first crossing. $\square$

The exact values $R(1) = 44/45$, $R(2) = 168/169$, $R(3) = 8232/7921$, $R(4) = 2860/2601$ are immediate from the displayed formula, and $R(3000) = 267.28\dots$. Note that $t = 1$ is plain K_4 with all six weights uniform, so $R(1) = 44/45$ is the disjoint-pair ratio p_2/p_0^2 at $n = 4$ and agrees with Table 1.

Remark 4.5 (why this does not contradict Theorem A). The law $\max(U_1, \dots, U_t)$ arises naturally inside an i.i.d. model: if an edge is subdivided into a path of t edges with private internal vertices and i.i.d. uniform weights, then, after subtracting the constant total path weight, the minimum spanning tree of the quotient is Kruskal on the base graph with the path *maxima* as effective weights. So the family of Theorem C is realisable at the level of *macro-edges*. It is not realisable at the level of a single fixed edge, and the loss is exactly quantifiable. Let X, Y be the indicators that the two macro-edges lie in the quotient tree, with probabilities $\mathbf{P}(X) = p$ and $\mathbf{P}(Y) = q$, and let I_1, I_2 be one fixed constituent edge in each path, both paths of length t . A constituent edge is omitted exactly when its macro-edge is omitted *and* it carries the maximum of its own path; the identity of that maximum is uniform along the path and independent of its value, so

$$\mathbf{P}(I_1) = 1 - \frac{1-p}{t} =: \alpha, \quad \mathbf{P}(I_2) = 1 - \frac{1-q}{t} =: \beta, \quad \text{Cov}(I_1, I_2) = \frac{\text{Cov}(X, Y)}{t^2},$$

for all p, q . If $p, q > 0$ and the macro-edge relative excess is nonzero, this identity may be divided by that excess to give

$$\frac{\mathbf{P}(I_1, I_2)/(\mathbf{P}(I_1)\mathbf{P}(I_2)) - 1}{\mathbf{P}(X, Y)/(pq) - 1} = \frac{pq}{t^2\alpha\beta} \leq \frac{1}{t^2},$$

the last step because $\alpha \geq p$ and $\beta \geq q$. Applied to (9) this leaves the fixed-edge ratio $1 + 5t^{-5} + O(t^{-6})$, so the amplification is destroyed. Theorem C is a theorem about the independent non-identical model, and Theorem A is a theorem about the i.i.d. model; the distinction is essential.

5. THE COMPLETE GRAPH

5.1. The accepted-merger coalescent. Give every edge of K_n an independent rate-one exponential clock and run Kruskal's algorithm in the order in which the clocks ring. This

does not change $\mathbf{P}_{\text{MST}}$, since the tree depends only on the weight order, and it has the advantage that the weights are now the ring times, so the total weight of the tree is L_n .

Discard the rejected clocks and record only the *accepted mergers*. There are exactly $n - 1$ of them, and they merge the vertex set from n singletons to one block.

Lemma 5.1. *At time zero, or just after any accepted merger other than the final one, condition on the whole past and let $A_1, \dots, A_k$ be the current components, of sizes $a_1, \dots, a_k$; thus $k \geq 2$. Then the next accepted merger joins A_i and A_j with probability*

$$(10) \quad \frac{a_i a_j}{Z}, \quad Z = \sum_{i < j} a_i a_j = \frac{n^2 - \sum_i a_i^2}{2},$$

and, conditionally on that pair, the two endpoints of the accepted edge are independent and uniform in A_i and A_j respectively, and independent of the past.

Proof. An edge of K_n whose endpoints lie in two distinct current components has not yet rung: had it rung earlier, Kruskal would have accepted it at that time and merged those components. By memorylessness of the exponential the residual clocks of all such edges are i.i.d. $\text{Exp}(1)$ given the past, so the next accepted edge is uniform among the Z cross edges. Uniformity among the $a_i a_j$ edges between A_i and A_j is exactly the assertion that, given the pair, the endpoints are independent and uniform; and this is independent of the past, which involves only clocks that have already rung. $\square$

Lemma 5.1 identifies the sequence of component sizes with the discrete skeleton of the multiplicative coalescent, cf. [2]. This is the only place where the complete graph enters. Direct Kruskal enumerations on small complete graphs agree with this transition model.

The two functionals below are the bookkeeping devices for the whole section.

Definition 5.2. A *merger history* is the sequence of pairs of component sizes merged, in order. For a merger history on n leaves put

$$H = \sum_{\text{mergers}} \left(\frac{1}{a} + \frac{1}{b} \right), \quad J = \sum_{\text{mergers}} \frac{1}{a + b},$$

where a, b are the sizes of the two components merged (the *children*) and $a + b$ is the size of the component created (the *parent*).

Lemma 5.3 (pathwise accounting). *For every merger history on n leaves,*

$$H = n - \frac{1}{n} + J.$$

Proof. Every component other than the final one of size n is a child of exactly one merger, so $H = \sum_{C \neq \text{root}} 1/|C|$, the sum being over all components ever present. The components ever present are the n singletons together with the $n - 1$ components created by mergers, and J sums $1/|C|$ over the created ones. Hence

$$H = \left(n + J \right) - \frac{1}{n},$$

the term n from the singletons and the correction $-1/n$ from removing the root, which is created but is nobody's child. $\square$

5.2. The degree functional. Let the accepted forest carry the degree function $\deg$ and set

$$\Phi = \sum_{x \in V} \deg(x)(\deg(x) - 1),$$

so that in the final tree Φ counts the ordered pairs of distinct tree edges sharing a vertex.

Lemma 5.4. *Conditionally on the entire merger history,*

$$(11) \quad \mathbf{E}[\Phi \mid \text{merger history}] = 8(n - 1) - 4H.$$

Proof. If a merger joins components of sizes a, b through endpoints u, v , then Φ increases by $[(\deg(u) + 1)\deg(u) - \deg(u)(\deg(u) - 1)] + [\text{same at } v] = 2\deg(u) + 2\deg(v)$. Each component is a tree, so the degrees inside a component of size a sum to $2(a - 1)$; by Lemma 5.1 the endpoint u is uniform in its component and independent of everything else, whence $\mathbf{E}[\deg(u) \mid \cdot] = 2(a - 1)/a$. Therefore

$$\mathbf{E}[\Delta\Phi \mid \cdot] = \frac{4(a - 1)}{a} + \frac{4(b - 1)}{b} = 8 - \frac{4}{a} - \frac{4}{b}.$$

The endpoint choices are independent of the sequence of component sizes, since the evolution (10) depends on sizes only; so we may condition on the whole merger history and sum over the $n - 1$ mergers, obtaining $8(n - 1) - 4H$ by Definition 5.2. $\square$

Identity (11) reduces both pair probabilities on K_n to the single functional H , and by Lemma 5.3 to J . Adjacent negative correlation needs J bounded below, disjoint negative correlation needs it bounded above, and the two directions require quite different handles on the same quantity.

5.3. Adjacent pairs: proof of Theorem B1. Write $H_n = \sum_{k=1}^n 1/k$ for the n th harmonic number.

Lemma 5.5 (chronological reciprocal bound). *Every merger history on n leaves satisfies*

$$(12) \quad J \geq H_n - 1.$$

Equality holds exactly for a comb history: after the first merger, each merger attaches one singleton to the component created at the preceding step.

Proof. Order the mergers chronologically. The component created by the j th merger has size at most $j + 1$, since a tree component on s vertices requires $s - 1$ accepted edges among those vertices. Therefore

$$J \geq \sum_{j=1}^{n-1} \frac{1}{j+1} = H_n - 1.$$

Equality requires the successive parent sizes to be $2, 3, \dots, n$. After the first merger, induction then forces the component just created to merge with a singleton at every step. Conversely such a comb history has precisely those parent sizes and attains equality. $\square$

Proof of Theorem B1. Lemmas 5.3 and 5.5 give $H \geq n - \frac{1}{n} + H_n - 1$ pathwise, and hence Lemma 5.4 gives

$$(13) \quad \mathbf{E}[\Phi] \leq 4 \left(n - 1 - H_n + \frac{1}{n} \right).$$

By vertex exchangeability, if D is the degree of a fixed vertex then $\mathbf{E}[D(D - 1)] = \mathbf{E}[\Phi]/n$. Finally, K_n has exactly $n(n - 1)(n - 2)$ ordered pairs of distinct edges sharing a vertex (choose the shared vertex, then an ordered pair of distinct other endpoints), and each contributes p_1 to $\mathbf{E}[\Phi]$. Consequently

$$\frac{p_1}{p_0^2} \leq \frac{n(n - 1 - H_n) + 1}{(n - 1)(n - 2)} < 1.$$

The last inequality is equivalent to $H_n > 2 - 1/n$, which holds at $n = 3$ and then by induction. Thus $p_1 < p_0^2$. $\square$

The chronological bound is sharp over merger histories, although not generally in expectation. At $n = 3$ it gives the exact ratio $p_1/p_0^2 = 3/4$.

5.4. Disjoint pairs: proof of Theorem B2. For the other direction we need J bounded above in expectation, and here the trivial estimate is useless: parent sizes can be as small as 2. The upper bound is where the multiplicative merger law (10) is used, and it comes from a convexity inequality applied one step at a time.

Lemma 5.6 (one-step partition inequality). *Let $k \geq 2$ and let $a_1, \dots, a_k$ be positive integers with $\sum_i a_i = n$. Then*

$$(14) \quad \sum_{i < j} \frac{a_i a_j}{a_i + a_j} \leq \frac{k}{2n} \sum_{i < j} a_i a_j.$$

Equality holds if and only if $k = 2$, or $a_1 = \dots = a_k$.

Proof. Write $S_2 = \sum_i a_i^2$ and $Z = \sum_{i < j} a_i a_j = (n^2 - S_2)/2$. From the identity

$$\frac{ab}{a+b} = \frac{a+b}{4} - \frac{(a-b)^2}{4(a+b)}$$

and from $a_i + a_j \leq n$ we get

$$\sum_{i < j} \frac{a_i a_j}{a_i + a_j} \leq \frac{1}{4} \sum_{i < j} (a_i + a_j) - \frac{1}{4n} \sum_{i < j} (a_i - a_j)^2.$$

Now $\sum_{i < j} (a_i + a_j) = (k-1)n$ and $\sum_{i < j} (a_i - a_j)^2 = (k-1)S_2 - 2Z = kS_2 - n^2$, so the right-hand side equals

$$\frac{(k-1)n}{4} - \frac{kS_2 - n^2}{4n} = \frac{k(n^2 - S_2)}{4n} = \frac{kZ}{2n},$$

which is (14).

For the equality case, the only inequality used was $(a_i - a_j)^2 / (a_i + a_j) \geq (a_i - a_j)^2 / n$, which is an equality for the pair $\{i, j\}$ exactly when $a_i = a_j$ or $a_i + a_j = n$. If $k = 2$ then $a_1 + a_2 = n$ and equality holds. If $k \geq 3$ and two parts differ, say $a_i \neq a_j$, then equality forces $a_i + a_j = n$, which is impossible since the remaining $k-2 \geq 1$ parts are positive. Hence for $k \geq 3$ equality holds exactly when all parts are equal, and in that case both sides equal $\binom{k}{2}n/(2k)$. $\square$

Since the left side of (14) is, up to the normalisation Z , exactly the conditional expectation of the increment of J , summing over the states visited turns it into the bound we need.

Lemma 5.7 (reciprocal budget). *For the accepted-merger coalescent on K_n ,*

$$(15) \quad \mathbf{E}[J] \leq \frac{(n-1)(n+2)}{4n}, \quad \text{hence} \quad \mathbf{E}[H] \leq \frac{(n-1)(5n+6)}{4n},$$

with strict inequality in both for $n \geq 4$, and equality for $n = 2, 3$.

Proof. By Lemma 5.1, from a state with component sizes $a_1, \dots, a_k$ the increment of J has conditional expectation

$$\mathbf{E}[\Delta J \mid \text{state}] = \frac{1}{Z} \sum_{i < j} \frac{a_i a_j}{a_i + a_j} \leq \frac{k}{2n}$$

by Lemma 5.6. The process visits a state with exactly k components once for each $k = n, n-1, \dots, 2$, and J is the sum of the increments at those states, so by the tower property

$$\mathbf{E}[J] \leq \sum_{k=2}^n \frac{k}{2n} = \frac{n(n+1)/2 - 1}{2n} = \frac{(n-1)(n+2)}{4n}.$$

The second bound in (15) follows from Lemma 5.3:

$$\mathbf{E}[H] = n - \frac{1}{n} + \mathbf{E}[J] \leq \frac{4n^2 - 4 + (n-1)(n+2)}{4n} = \frac{(n-1)(5n+6)}{4n}.$$

For strictness, note that for $n \geq 4$ the state reached after the first merger is $(2, 1, 1, \dots, 1)$ with $k = n - 1 \geq 3$ components which are not all equal, so Lemma 5.6 is strict there. For $n = 2$ and $n = 3$ every state visited has $k = 2$ or is the all-ones state, so every step is an equality; indeed $J \equiv 5/6$ and $H \equiv 7/2$ deterministically at $n = 3$. $\square$

Remark 5.8 (the merger law is needed, not only the state space). Unlike (12), Lemma 5.7 has no pathwise version. For the balanced merger history on $n = 4$ leaves — merge two disjoint pairs of singletons, then merge the two blocks — we have

$$J = \frac{1}{2} + \frac{1}{2} + \frac{1}{4} = \frac{5}{4} > \frac{(n-1)(n+2)}{4n} = \frac{9}{8},$$

so by (11) that history has $\mathbf{E}[\Phi \mid \text{history}] = 4$, below the threshold $3(n-1)(n-2)/n = 4.5$ that the proof below requires. Nor would the state space alone suffice: replacing the multiplicative choice (10) by a uniform choice among the $\binom{k}{2}$ pairs of blocks breaks Lemma 5.6 at once. At the state $(1, 1, 2)$, where $n = 4$, $k = 3$ and the bound $k/(2n)$ is $3/8$, the multiplicatively weighted average $Z^{-1} \sum_{i < j} a_i a_j / (a_i + a_j)$ equals $11/30 \leq 3/8$, while the unweighted average of $1/(a_i + a_j)$ is $7/18 > 3/8$.

Proof of Theorem B2. Let $n \geq 4$. Combining (11) with Lemma 5.7,

$$(16) \quad \mathbf{E}[\Phi] = 8(n-1) - 4\mathbf{E}[H] > 8(n-1) - \frac{(n-1)(5n+6)}{n} = \frac{3(n-1)(n-2)}{n}.$$

The tree has $(n-1)(n-2)$ ordered pairs of distinct edges, of which Φ are adjacent; hence it has $(n-1)(n-2) - \Phi$ ordered disjoint pairs. The number of ordered pairs of disjoint edges of K_n is $6\binom{n}{4} = n(n-1)(n-2)(n-3)/4$, and each contributes p_2 , so

$$p_2 = \frac{4[(n-1)(n-2) - \mathbf{E}[\Phi]]}{n(n-1)(n-2)(n-3)} < \frac{4(n-1)(n-2)[1 - 3/n]}{n(n-1)(n-2)(n-3)} = \frac{4(n-3)}{n^2(n-3)} = \frac{4}{n^2} = p_0^2. \quad \square$$

Proof of Corollary B3. Two distinct edges of K_n either share a vertex or not; apply Theorem B1 or Theorem B2. For $n = 3$ no disjoint pair exists. $\square$

6. THE IDENTITY, AND THE REDUCTION TO $\mathbf{E}[L_n]$

The bounds of §5 were two estimates of the same functional H . That functional has an exact meaning, and identifying it is what turns the pair correlations on K_n into a statement about a classical quantity. Recall that L_n is the weight of $\text{MST}(K_n)$ under i.i.d. $\text{Exp}(1)$ weights, so that in the coalescent of §5 the weights are the clock ring times.

Lemma 6.1. $\mathbf{E}[H] = n\mathbf{E}[L_n]$ for every $n \geq 2$.

Proof. Let $\kappa(t)$ be the number of components of the graph formed by the edges of weight at most t . Since the weight of an edge is its ring time,

$$L_n = \sum_{g \in T} w(g) = \int_0^\infty \#\{g \in T : w(g) > t\} dt = \int_0^\infty (\kappa(t) - 1) dt,$$

because at time t the number of accepted mergers still to come is exactly $\kappa(t) - 1$. In the continuous-time process, at a state with components of sizes $a_1, \dots, a_k$ the total accepted-merger rate is Z and the expected rate at which the functional H accumulates is

$$\sum_{i < j} a_i a_j \left(\frac{1}{a_i} + \frac{1}{a_j} \right) = \sum_{i < j} (a_i + a_j) = n(k-1),$$

while $\kappa - 1 = k - 1$. Thus $H_t - n \int_0^t (\kappa(s) - 1) ds$ is a martingale. Since $H \leq 2(n-1)$ and L_n is integrable, letting $t \rightarrow \infty$ gives $\mathbf{E}[H] = n\mathbf{E}[L_n]$. $\square$

n	p_1/p_0^2	p_2/p_0^2	$\mathbf{E}[L_n]$	harmonic lower bd.	upper bd.	$\frac{(n-1)(5n+6)}{4n^2}$
3	0.7500000000	—	1.1667	1.1667	7/6	
4	0.7555555556	0.9777777778	1.2167	1.2083	39/32	
5	0.7597552910	0.9804894180	1.2353	1.2167	31/25	
6	0.7630719281	0.9825707626	1.2427	1.2139	5/4	
7	0.7657742674	0.9842257326	1.2454	1.2071	123/98	
8	0.7680285888	0.9855771290	1.2460	1.1991	161/128	
9	0.7699442296	0.9867038469	1.2455	1.1909	34/27	
10	0.7715965139	0.9876591349	1.2445	1.1829	63/50	
11	0.7730392810	0.9884803595	1.2432	1.1754	305/242	
12	0.7743121839	0.9891945849	1.2418	1.1683	121/96	
13	0.7754451621	0.9898219351	1.2405	1.1618	213/169	
14	0.7764613009	0.9903777088	1.2391	1.1557	247/196	
20	0.7809063153	0.9927279258	1.2323	1.1274	1007/800	
30	0.7850808025	0.9948028441	1.2250	1.0987	377/300	

TABLE 1. Exact pair correlations for $\text{MST}(K_n)$, the harmonic lower bound, and the upper bound from Corollary D2. Every entry comes from an exact rational; the correlation-ratio, $\mathbf{E}[L_n]$, and harmonic-bound decimals are rounded as displayed. The full exact table for $3 \leq n \leq 30$ accompanies the paper.

Proof of Proposition D1. With D the degree of a fixed vertex, $\mathbf{E}[D(D-1)] = \mathbf{E}[\Phi]/n$ by exchangeability and $\mathbf{E}[D] = 2(n-1)/n$ because every spanning tree has $n-1$ edges. Hence, by (11) and Lemma 6.1,

$$\mathbf{E}[D^2] = \frac{\mathbf{E}[\Phi]}{n} + \frac{2(n-1)}{n} = \frac{8(n-1) - 4n\mathbf{E}[L_n]}{n} + \frac{2(n-1)}{n} = \frac{10(n-1)}{n} - 4\mathbf{E}[L_n]. \quad \square$$

Proof of Corollary D2. For $n \geq 3$, Proposition D1 and (11) show that $p_1 \leq p_0^2$ is $\mathbf{E}[\Phi] \leq 4(n-1)(n-2)/n$, i.e. $8(n-1) - 4n\mathbf{E}[L_n] \leq 4(n-1)(n-2)/n$, i.e. $\mathbf{E}[L_n] \geq (n-1)(n+2)/n^2$. For $n \geq 4$, likewise $p_2 \leq p_0^2$ is $\mathbf{E}[\Phi] \geq 3(n-1)(n-2)/n$, i.e. $\mathbf{E}[L_n] \leq (n-1)(5n+6)/(4n^2)$. Lemma 5.5, Lemma 5.3, and Lemma 6.1 give

$$\mathbf{E}[L_n] \geq 1 + \frac{H_n - 1}{n} - \frac{1}{n^2}.$$

This implies the weaker lower threshold in the equivalence because $H_n - 1 \geq (n-1)/n$. Equality in the harmonic bound holds for $n = 2, 3$; for $n \geq 4$ a non-comb history has positive probability, so it is strict. The upper estimate and its equality statements follow from Lemma 5.7. $\square$

Thus the two pair inequalities become bounds on $\mathbf{E}[L_n]$, proved here without input from the minimum-spanning-tree-weight literature. The harmonic lower bound still tends to 1 and is asymptotically loose. The upper bound tends to 5/4: it is exact at $n = 2, 3$, within 1% of $\mathbf{E}[L_n]$ through $n = 10$, and about 4% above the limit $\zeta(3)$. See Table 1. Both pair ratios increase in the computed range. The maximum of $\mathbf{E}[L_n]$ in Gamarnik's range $2 \leq n \leq 45$ occurs at $n = 8$, where $\mathbf{E}[L_8] = 1.2459812\dots$ against the upper bound $161/128 = 1.2578125$; decrease for every $n \geq 8$ remains conjectural [10, Conjecture 4.1].

Proposition D1 also recasts Gamarnik's conjectured decrease: for $n \geq 8$, $\mathbf{E}[L_n] > \mathbf{E}[L_{n+1}]$ is equivalent to

$$\mathbf{E}[\deg_{n+1}(x)^2] - \mathbf{E}[\deg_n(x)^2] > \frac{10}{n(n+1)},$$

where $\deg_k$ is the degree in $\text{MST}(K_k)$.

Remark 6.2 (what Theorem B2 says about the expected tree weight). Corollary D2 exposes Theorem B2 as the non-asymptotic bound

$$(17) \quad \mathbf{E}[L_n] \leq \frac{(n-1)(5n+6)}{4n^2} \quad (n \geq 2),$$

a statement about a quantity with a long history: Frieze [9], Steele [18], Janson [12], Fill and Steele [8], Gamarnik [10], Nishikawa, Otto and Starr [17], Cooper, Frieze, Ince, Janson and Spencer [6], and Li and Zhang [13]. These works develop asymptotics, exact small- n evaluations, and polynomial or Tutte representations. None of these sources states the all- n upper bound (17). It is exact at $n = 2, 3$ and lies within 0.0119 of the largest value in Gamarnik's computed range, but its elementary form makes prior knowledge plausible; Theorem B2 may therefore be a new complete-range proof of a known bound rather than a new inequality. Theorem B1 uses the other half of Corollary D2 and is unaffected.

Proof of Corollary D3. The expansion of Cooper, Frieze, Ince, Janson and Spencer [6] in particular gives $\mathbf{E}[L_n^{\text{unif}}] \rightarrow \zeta(3)$ for i.i.d. Uniform $[0, 1]$ weights. Li and Zhang [13] give the quantitative transfer

$$(18) \quad \mathbf{E}[L_n] - \mathbf{E}[L_n^{\text{unif}}] = \frac{\zeta(3)}{n} + O\left(\frac{\log^2 n}{n^2}\right),$$

so $\mathbf{E}[L_n] \rightarrow \zeta(3)$ for exponential weights as well. Insert this into Proposition D1:

$$\lim_{n \rightarrow \infty} \mathbf{E}_{\text{MST}(K_n)}[\deg(x)^2] = 10 - 4\zeta(3) = 5.1917723873616 \dots$$

By [19, Proposition 3.11] this is exactly $\mathbf{E}[N^2]$ for the root degree N of the wired minimal spanning forest on the Poisson-weighted infinite tree [16], which is the assertion of [19, Question 3.8]. $\square$

The sharper expansion identifies the rates and the two limiting correlation ratios. Let c_1, c_2 be the constants in [6, Theorem 1], so that

$$(19) \quad \mathbf{E}[L_n^{\text{unif}}] = \zeta(3) + \frac{c_1}{n} + \frac{c_2 + o(1)}{n^{4/3}}, \quad c_1 = 0.0384956 \dots$$

Corollary 6.3 (asymptotic rates). *For exponential weights,*

$$\begin{aligned} \mathbf{E}[L_n] &= \zeta(3) + \frac{c_1 + \zeta(3)}{n} + \frac{c_2 + o(1)}{n^{4/3}}, \\ \mathbf{E}[\deg(x)^2] &= 10 - 4\zeta(3) - \frac{10 + 4(c_1 + \zeta(3))}{n} - \frac{4c_2 + o(1)}{n^{4/3}}. \end{aligned}$$

Moreover,

$$\begin{aligned} \frac{p_1}{p_0^2} &= 2 - \zeta(3) + \frac{4 - c_1 - 4\zeta(3)}{n} - \frac{c_2 + o(1)}{n^{4/3}}, \\ \frac{p_2}{p_0^2} &= 1 + \frac{4\zeta(3) - 5}{n} + O(n^{-2}). \end{aligned}$$

In particular the degree second moment converges at order n^{-1} , while the adjacent and disjoint ratios tend to $2 - \zeta(3)$ and 1, respectively.

Proof. The first line follows by combining (18) and (19); Proposition D1 gives the second. If $m_n = \mathbf{E}[\deg(x)^2]$, direct counting of ordered adjacent and disjoint edge pairs gives

$$\frac{p_1}{p_0^2} = \frac{m_n - 2 + 2/n}{4(1 - 3/n + 2/n^2)} \quad (n \geq 3), \quad \frac{p_2}{p_0^2} = \frac{n^2((n-1) - m_n)}{(n-1)(n-2)(n-3)} \quad (n \geq 4).$$

Substitution and expansion prove the last two formulas. $\square$

This argument does not pass through the local weak limit. The local weak limit of $\text{MST}(K_n)$ is due to Addario-Berry [1], and convergence in distribution of the root degree does not by itself give convergence of the second moment, which is why Question 3.8 was asked; the recent study of dynamic local convergence for Prim's algorithm [7] likewise does not address moments. Proposition D1 replaces the missing uniform integrability by an exact finite identity. Corollary 6.3 gives an asymptotic rate, although the $o(1)$ in (19) does not supply explicit finite error constants. Tang and Zhang [19, Proposition 3.11] print the decimal 5.191717; the correct value of $10 - 4\zeta(3)$ is 5.1917723873616..., and the closed form there is not affected.

7. OPEN PROBLEMS

Theorem A locates positive correlation in the covariance of two bottleneck distances, while Proposition D1 controls the complete-graph pair correlations through the expected tree weight. Both descriptions leave natural quantitative questions.

Problem 7.1 (the optimal constant). Determine the least C for which the conclusion of Theorem A holds. We know only that it lies in $[13938405/13872419, 8]$. Corollary 3.5 gives the sharper factor $2 + o(1)$ when both marginals vanish, but does not establish a global factor 2. The factor 8 comes from the sharp inequality $\mathbf{E}[T^2] \leq 2(\mathbf{E}T)^2$ and one factor 2 from each deletion sandwich; improving it requires using more of the joint structure than these three separate bounds.

Problem 7.2 (the simple-graph constant). Determine the supremum of $p_{ef}/(p_e p_f)$ over finite connected simple graphs. Proposition 4.4 gives the lower bound $13938405/13872419$, while Theorem A gives the upper bound 8.

Problem 7.3 (effective errors and monotonicity). Corollary 6.3 gives the asymptotic rate and the limits of both complete-graph ratios, but not explicit finite error constants. Can the error in (19) be made effective enough to give useful finite- n bounds? Both ratios increase for $3 \leq n \leq 30$ where defined (Table 1); are they monotone throughout their domains?

Problem 7.4 (other families at every size). Tang and Zhang [19, Remark 1.2] state that their method also gives the disjoint-pair property for $K_{n,n}$ for all sufficiently large n . Is there an all- n analogue of Corollary B3 for $K_{n,n}$, or for other natural families? The method of §5 uses the complete-graph transition rate (10) and does not transfer as it stands. The minimum spanning tree weight has been studied on several other host graphs, for instance regular graphs [4], so the analogue of Corollary D2 there is a concrete starting point. Finally, is there an analogue of Theorem A for the wired minimal spanning forest on an infinite graph?

APPENDIX A. EXACT COMPUTATIONS AND COMPUTER ASSISTANCE

The universal statements of §3 and the complete-graph theorems of §5–6 use no computation. Computation supplies the exact table and the finite enumerations in Propositions 4.3 and 4.4, and checks the algebra of Proposition 4.2 and the coalescent model against direct evaluations of $\mathbf{P}_{\text{MST}}$. Every computed quantity is an exact rational; decimals are rounded only when printed. The full table and standard-library Python code are available in the public companion repository <https://github.com/agupta/random-mst-correlations>; the citable preprint is archived at [doi:10.5281/zenodo.21780630](https://doi.org/10.5281/zenodo.21780630).

A.1. What each computation evaluates.

- (1) *Direct Kruskal enumerations.* One route enumerates all $m!$ edge orders; another sums exactly over the relative-order blocks determined by the marked weights. These are distinct finite enumerations, but both use the same Kruskal/union–find semantics. They cover K_4 , K_5 , the small-graph examples, and the two- and three-hub instances; a symmetry-compressed version handles the hub family for $1 \leq s \leq 9$.
- (2) *The accepted-merger coalescent.* A dynamic program over integer partitions with transition probabilities (10) returns $\mathbf{E}[J]$, $\mathbf{E}[H]$, $\mathbf{E}[\Phi]$, $\mathbf{E}[D^2]$, p_1 , p_2 and $\mathbf{E}[L_n]$ exactly. This implements the proof in §5, rather than providing independent evidence for it; the direct Kruskal computations test it in small cases.
- (3) *$\mathbf{E}[L_n]$ from the area identity.* Independently of the coalescent, the component count $\kappa(t)$ of $G(n, 1 - e^{-t})$ gives

$$\mathbf{E}[L_n] = \int_0^\infty (\mathbf{E}[\kappa(t)] - 1) dt.$$

Substituting $u = e^{-t}$ turns the integrand into an integer polynomial in u built from the exact connectivity polynomial of $G(s, p)$, and the integral becomes a finite rational sum. This route uses neither mergers nor degrees.

The direct routes agree on every marginal and pair probability of K_4 and agree with the coalescent program on p_1 , p_2 , and $\mathbf{E}[\Phi] = 8(n-1) - 4\mathbf{E}[H]$ for $n = 4, 5$. The connectivity-polynomial route agrees with it on $\mathbf{E}[L_n]$, thereby checking Proposition D1 without reusing the coalescent identity $\mathbf{E}[H] = n\mathbf{E}[L_n]$. The resulting table agrees with [10] through $n = 30$; the committed transcription extends through $n = 45$. Internal identities such as $p_0 = 2/n$, $\sum_e p_e = n-1$, and $\mathbf{E}[H] = n - \frac{1}{n} + \mathbf{E}[J]$ are also checked, but are not counted as independent routes.

A.2. What is verified, and in what range. The documented exhaustive ranges are as follows. Merger histories are checked through $n = 12$, integer partitions through $n = 32$, and the closing rational algebra through $n < 300$. Theorem A and its frozen-environment lemmas are checked on every connected labelled multigraph with at most four vertices and at most six edges, allowing loops, parallel classes, and disconnected deleted environments. The bundle certificates are re-expanded coefficientwise and the family is evaluated for $r+s \leq 120$; the $R(t+1) > R(t)$ certificate is similarly re-expanded and scanned through $t = 300$. The hub-family computation covers $1 \leq s \leq 9$. The simple-graph minimality assertion uses the bridge reduction in the text and a census of 1528 labelled bridgeless graphs with at most seven edges, reducing to 17 isomorphism classes and 272 representative marked pairs. Theorem C and Remark 4.5 are also checked directly for subdivision lengths $t = 1, 2, 3$.

Several tempting strengthenings fail, and the suite retains them as controls: $C = 1$ fails by (1); $\mathbf{E}[T^2] \leq \frac{3}{2}(\mathbf{E}T)^2$ fails for the minimum of six uniforms; $a \geq \frac{3}{4}A$ fails when the marked edges join the same two vertices and no other edge is present; Lemma 3.3 is an equality when both marked edges are bridges; and Lemma 5.6 fails at $(2, 1, 1)$ if block pairs are chosen uniformly rather than with multiplicative weights.

These checks establish only the stated finite assertions and test the modelling and algebra; they do not replace the analytic proofs. Complete commands and ranges are recorded in the companion repository README.

ACKNOWLEDGMENT OF GENERATIVE-AI ASSISTANCE

Anthropic Claude Code (Claude 5 family) and OpenAI Codex (GPT-5.6 family) were used extensively for proof exploration, software development, exact computational checks, literature discovery, and drafting and editing the manuscript. The author selected the arguments and methods, checked the cited sources and reported computations, and takes full responsibility for the content. These systems are not authors or independent guarantors of correctness.

DECLARATIONS

The author declares no competing interests. This research received no external funding. The manuscript and accompanying data are released under CC BY 4.0; the accompanying software is released under the MIT License.

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